

Exploratory Data Analysis of the Road-Segment Risk Dataset

ROAD-SEGMENT RISK CLASSIFICATION USING DECISION TREES AND LIGHTGBM

1. Introduction

Exploratory Data Analysis (EDA) is performed to understand the structure, quality, distribution, relationships, and characteristics of the road-segment risk dataset before applying machine-learning algorithms. The analysis helps identify missing values, duplicate records, class imbalance, unusual observations, feature distributions, and relationships between roadway characteristics and the target risk category.

The dataset is intended for a supervised multiclass classification problem in which individual road segments are classified into risk categories. Depending on the selected dataset, the target may be represented as Low Risk, Medium Risk, and High Risk, or another documented set of categories. The purpose of EDA is to understand the available variables and verify that they are suitable for the proposed Decision Tree and LightGBM models.

The analysis follows the organization and level of detail of the supplied EDA report, but all discussion is adapted to the road-segment risk-classification topic.

2. Dataset Overview

The road-risk dataset contains observations representing individual road segments. Each row corresponds to a segment or road location, while the columns describe roadway, traffic, infrastructure, environmental, and historical safety characteristics. The exact number of records and columns depends on the selected dataset and should be reported from the actual file used for the project.

Column / Feature Group	Data Type	Description	Role
segment_id	Object/Integer	Unique road-segment identifier	Identifier
traffic_volume	Numeric	Traffic exposure or average vehicle volume	Input
speed_limit	Numeric	Posted speed limit or speed environment	Input
lanes	Numeric	Number of traffic lanes	Input
road_width	Numeric	Approximate roadway width	Input
intersection_density	Numeric	Number/density of intersections	Input
pavement_condition	Numeric/Categorical	Road-surface condition	Input
lighting	Categorical	Lighting condition	Input
weather	Categorical/Numeric	Weather or environmental condition	Input
historical_crashes	Numeric	Historical crash measure	Input
risk_class	Categorical	Low/Medium/High or project-defined class	Target

The table is a project-oriented schema. Only variables actually present in the chosen road dataset should be retained in the final report. If the dataset uses different names, those names should be reported exactly as they appear in the file.

3. Dataset Structure and Data Types

The first EDA step is to inspect the number of observations, number of variables, data types, and non-null counts. Numerical variables such as traffic volume, speed, road width, lane count, and crash frequency should be represented using appropriate numeric types. Categorical variables such as lighting, pavement condition, road type, weather, or intersection type should be treated as categorical information.

The target `risk_class` must be clearly separated from the input features. An identifier such as `segment_id` should generally not be used as a predictive feature because it identifies the record rather than describing road conditions. Geographic coordinates may be retained when they have a justified spatial role, but they should be handled carefully because they can cause the model to learn location-specific patterns.

Observation: Correct data types are important because Decision Tree and LightGBM models require machine-readable feature values. Categorical variables may require encoding depending on the implementation and the chosen LightGBM configuration.

4. Missing Value Analysis

Missing values are examined for every column. The number and percentage of missing observations should be calculated before model training. Missing traffic counts, road-condition values, weather observations, or crash-history values may occur because different data sources are collected at different times.

If missing values are present, they should be handled using a documented strategy. Numerical variables can be imputed using a training-set statistic such as the median, while categorical variables can use the most frequent category or an explicit Unknown category. The preprocessing parameters must be learned from the training data to avoid data leakage.

Observation: If the actual dataset contains zero missing values, this should be reported directly. If missing values exist, they should not simply be deleted without checking how many observations would be lost and whether missingness is associated with particular road types or risk classes.

5. Duplicate Data Analysis

Duplicate records are checked using a duplicate-row operation. Duplicate road-segment observations can occur when data from multiple sources are combined or when the same segment is recorded more than once. Exact duplicates should normally be investigated before modelling.

The analysis should report total duplicate rows and duplicate percentage. If duplicates are found, the project should determine whether they represent accidental repetition or legitimate repeated measurements from different dates. Time-stamped observations should not automatically be treated as duplicates because the same road segment can legitimately appear in multiple time periods.

Observation: Removing true accidental duplicates prevents a particular road segment from receiving excessive influence during model training.

6. Target Variable Analysis

The target variable is `risk_class`. The dataset should be examined to determine how many road segments belong to each risk category. A typical formulation for this project is three-class classification: Low Risk, Medium Risk, and High Risk.

Risk Class	Number of Segments	Percentage
Low Risk	Use actual count	Calculate from dataset
Medium Risk	Use actual count	Calculate from dataset
High Risk	Use actual count	Calculate from dataset
Total	Use actual count	100%

Observation: The class distribution is important because a dataset with substantially more Low-risk segments than High-risk segments is imbalanced. Accuracy alone may then give an overly optimistic impression of model performance. Precision, recall, F1-score, and the confusion matrix should therefore be included in later evaluation.

7. Numerical Feature Analysis

Numerical features are analyzed using descriptive statistics such as count, mean, standard deviation, minimum, quartiles, median, and maximum. For this road-risk project, important numerical variables may include traffic volume, speed limit, road width, number of lanes, intersection density, curvature, pavement score, and historical crash frequency.

Descriptive statistics help determine whether the variables have reasonable ranges. For example, lane count should not contain negative values, traffic volume should use a documented unit, and speed values should be consistent across observations. Extremely large values should be investigated rather than automatically deleted.

The difference between mean and median can indicate skewness. Traffic volume and historical crash counts may be strongly right-skewed because a small number of road segments can have much higher exposure than the majority of segments. Such patterns should be visualized before selecting transformations or thresholds.

8. Roadway Feature Analysis

Roadway characteristics are examined to understand how infrastructure varies across the network. Features such as lane count, road width, road type, curvature, shoulder condition, median type, pavement condition, and intersection density can provide information about the physical environment of each segment.

Categorical distributions can be summarized using frequency tables and bar charts. For example, the number of segments with good, moderate, and poor pavement condition can be compared. Similar analysis can be performed for lighting condition, road type, intersection type, and surface type.

The objective is not to assume that a particular feature causes risk, but to identify patterns that may be useful to the machine-learning models and to detect unusual or inconsistent records.

9. Traffic Feature Analysis

Traffic-related variables are especially relevant to road-segment risk classification. Traffic volume describes exposure, while speed-related variables describe the operating environment. Heavy-vehicle percentage, congestion indicators, peak-hour traffic, and traffic density may provide additional information when available.

Traffic variables should be checked for unit consistency and plausible ranges. If traffic counts are collected at different times of day, the time period represented by each value should be documented. Comparing peak-hour and daily measures without appropriate interpretation can produce misleading relationships.

Group comparisons can be performed by risk class. For example, average traffic volume can be calculated separately for Low-, Medium-, and High-risk segments. Similar comparisons can be made for speed limit, heavy-vehicle ratio, and congestion indicators. These comparisons provide useful EDA insights before model training.

10. Historical Crash Feature Analysis

Historical crash information may be one of the strongest predictors of a risk label when the target is related to previous safety outcomes. Relevant variables can include crash count, severe-crash count, injury-crash count, fatal-crash count, or an exposure-adjusted crash rate.

Historical variables must be constructed carefully to avoid target leakage. If the model is intended to predict risk for a future period, only information available before that prediction period should be used as an input. Future crash outcomes must not be included among the predictors.

Observation: A high historical crash count may indicate a segment requiring attention, but it should not automatically be interpreted as proof that the same conditions will continue indefinitely. Roadway improvements and changing traffic conditions can alter future risk.

11. Outlier Detection

Potential outliers in numerical road features can be identified using the Interquartile Range (IQR) method. For a variable, Q1 is the first quartile and Q3 is the third quartile. The IQR is $Q3 - Q1$. Values below $Q1 - 1.5 \times \text{IQR}$ or above $Q3 + 1.5 \times \text{IQR}$ can be flagged as potential outliers.

For road-risk data, outliers may occur in traffic volume, road width, speed, crash counts, intersection density, or other measurements. A very high traffic count may be a genuine observation for a major highway rather than an error. Therefore, an outlier should be investigated using domain context before removal.

Outlier analysis is used primarily to understand the dataset. Removing all statistically unusual values can eliminate precisely the observations that are important for identifying high-risk road conditions.

12. Feature Distribution Analysis

Histograms and density-style plots can be used to understand the distribution of numerical variables. Traffic volume may show a long right tail, while speed limits may be concentrated around a smaller set of standard values. Crash counts may contain many zeros with a smaller number of segments having multiple recorded crashes.

Understanding distributions assists with feature engineering and interpretation. Tree-based models generally do not require standardization of numerical variables in the same way as many distance-based algorithms. Nevertheless, distribution analysis remains important for identifying data-quality problems and understanding the road network.

13. Feature Relationships and Correlation Analysis

Correlation analysis can be used to examine relationships among numerical variables. Traffic volume may be related to road width and lane count, while intersection density may be related to access-point density. Strong correlations indicate that multiple features may contain overlapping information.

Correlation does not establish causation. A feature may also act as a proxy for another variable that is not directly recorded. Feature importance from Decision Tree or LightGBM should therefore be interpreted together with domain knowledge and, where possible, additional explanation techniques.

For tree-based algorithms, highly correlated variables do not necessarily need to be removed. However, correlation can make feature-importance interpretation less straightforward because several related variables may share predictive information.

14. Risk Class vs Feature Analysis

One of the most useful EDA steps is comparing feature distributions across risk classes. Boxplots can compare traffic volume, speed, crash history, or road width among Low-, Medium-, and High-risk groups. Bar charts can compare categorical variables such as lighting or pavement condition across the target classes.

If High-risk segments show systematically different distributions for one or more variables, those features may be useful to the classifiers. However, overlap between classes is also important. Strong overlap indicates that no single variable can completely separate the categories and supports the use of multivariate machine learning.

15. Visualization Analysis

Several visualizations should be generated during EDA.

15.1 Risk-Class Distribution

A count plot or bar chart should display the number of Low-, Medium-, and High-risk segments. This visualization clearly shows whether the target is balanced or imbalanced.

15.2 Numerical Feature Histograms

Histograms can be created for traffic volume, speed limit, road width, lane count, intersection density, and historical crashes. These plots help identify skewness, concentration, and unusual observations.

15.3 Boxplots by Risk Class

Boxplots can compare important numerical variables across Low-, Medium-, and High-risk segments. They can reveal differences in medians, spread, and potential outliers.

15.4 Correlation Heatmap

A correlation heatmap can summarize relationships among numerical predictors. It is useful for identifying strongly related variables before modelling.

15.5 Categorical Feature Charts

Bar charts can show distributions of road type, pavement condition, lighting, weather category, and intersection type. Grouped or stacked charts can additionally compare these categories against the risk class.

16. Sample Road-Segment Inspection

Sample records should be inspected after the general EDA to verify that the dataset contains meaningful road-segment information. A sample may contain a segment identifier, traffic measurements, roadway geometry, infrastructure condition, environmental variables, historical safety information, and the assigned risk class.

Inspection should verify that units and category labels are consistent. For example, pavement condition should not contain several spellings for the same category, and speed should not combine different units without conversion. If latitude and longitude are available, a small sample can also be checked against the intended geographic region.

Observation: Sample inspection is a simple but important quality-control step because summary statistics alone may not reveal incorrectly entered values, inconsistent categories, or leakage from identifiers.

17. Overall EDA Findings

EDA Component	Finding to Report
Total records	Actual number from selected dataset
Input features	Actual feature count
Target variable	risk_class / project-defined target
Problem type	Multiclass classification if 3+ risk categories
Learning type	Supervised Machine Learning
Missing values	Actual count and percentage
Duplicate records	Actual count and percentage
Class distribution	Low / Medium / High counts and percentages
Important numerical features	Based on EDA
Potential outliers	Count by feature
Strong relationships	Based on correlation/group analysis

18. Implications for Decision Tree and LightGBM

The EDA findings directly guide the modelling stage. If the target classes are imbalanced, class-aware evaluation and possibly class weighting should be considered. If several features contain missing values, preprocessing must be included in the model pipeline. If numerical features contain extreme but valid values, they should be retained unless there is evidence of measurement error.

The Decision Tree can use the feature relationships discovered during EDA to construct interpretable splitting rules. Tree depth, minimum samples, and pruning should be controlled to reduce overfitting.

LightGBM can learn more complex nonlinear relationships and interactions among the roadway, traffic, environmental, and historical features. Hyperparameters should be tuned using validation data or cross-validation rather than the final test set.

The EDA stage therefore acts as the foundation for the proposed classification system. It ensures that the models are trained on understood and appropriately prepared data rather than raw observations with unknown quality problems.

19. Conclusion of EDA

The exploratory data analysis of the road-segment risk dataset provides the foundation for developing a supervised machine-learning system for ROAD-SEGMENT RISK CLASSIFICATION USING DECISION TREES AND LIGHTGBM. The analysis examines dataset structure, feature types, missing values, duplicates, target-class distribution, numerical features, roadway characteristics, traffic variables, historical safety information, outliers, correlations, and visual patterns.

The EDA should be used to document the actual characteristics of the selected dataset rather than assuming values that are not present. In particular, the number of road segments, class percentages, missing values, duplicate records, and outlier counts must be calculated from the final project dataset.

Overall, the EDA confirms whether the dataset is suitable for the next stage of the project. After preprocessing and feature preparation, the data can be used to train and compare a Decision Tree and LightGBM classifier. Model evaluation should include accuracy, precision, recall, F1-score, and confusion matrix, with special attention to the detection of High-risk road segments.

The next stage of the project will therefore focus on preprocessing, feature engineering, training the Decision Tree and LightGBM models, hyperparameter tuning, cross-validation, comparative evaluation, and interpretation of road-segment risk predictions.